



RESEARCH SEMINAR

Forecasting with Deep Temporal Hierarchies

Personal Details

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Main Resources:

1. Forecasting with Deep Temporal Hierarchies: A novel way for forecasting with temporal hierarchies based on deep learning models. - Master Thesis -
2. Theodosiou, Filotas & Kourentzes, Nikolas. (2021) Forecasting with Deep Temporal Hierarchies. SSRN Scholarly Paper ID 3918315. Social Science Research
3. Theodosiou, Filotas & Kourentzes, Nikolaos. (2021). Deep Learning Temporal Hierarchies for Interval Forecasts. International Conference of AI in Finance. Time Series Workshop



Content

- Why Temporal Hierarchies?
- Temporal Hierarchies in a Nutshell
- Non-Linear Data-Driven Reconciliation
 - Positional Information
 - Variance Scaling
 - Weight Optimization
- Experiment
- End-to-End Framework
- Prediction Intervals
- Conclusions

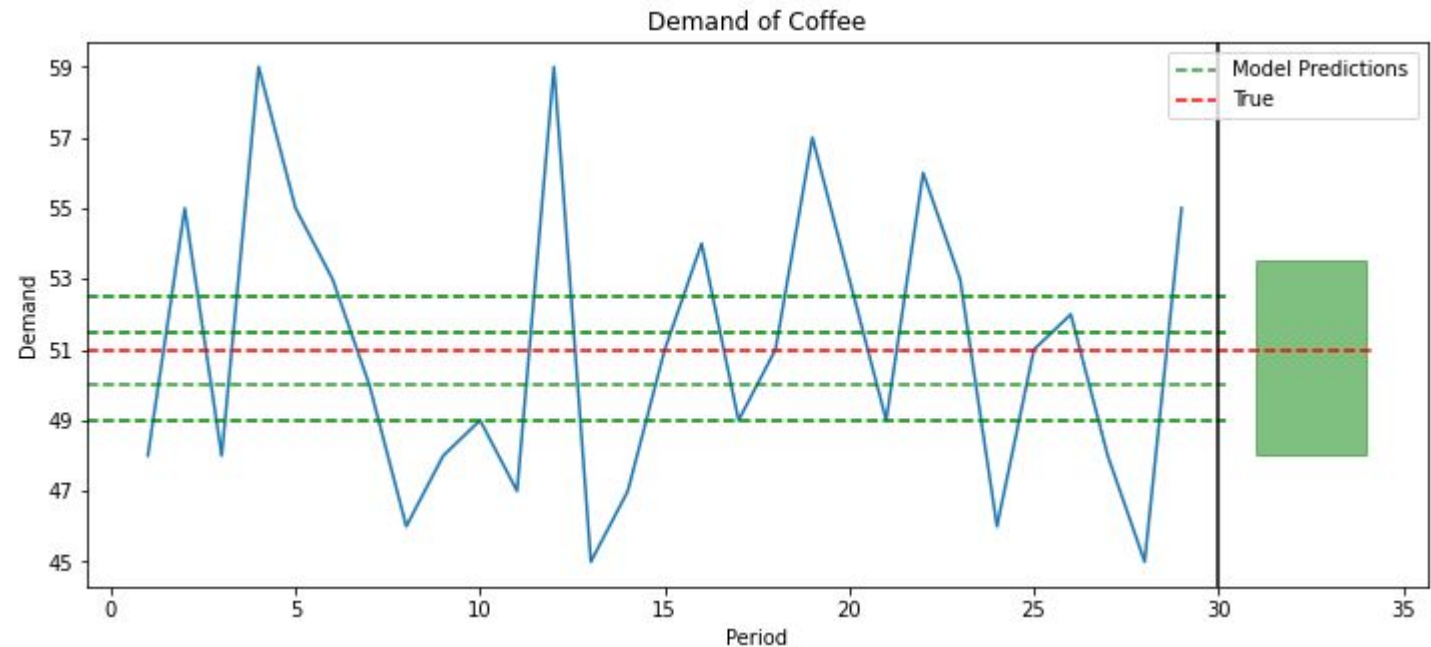
Why Temporal Hierarchies? Model Selection Uncertainty

We forecast the demand of coffee.

- We assume demand is independent of weather/price/location
- Constant demand with mean = 51 with some noise

We generate predictions using different models
But we get high forecast uncertainty

- Model selection uncertainty
- Parametric uncertainty
- Forecast uncertainty originated from noise



Why Temporal Hierarchies? Forecast Combinations

In every forecasting problem we look for a “good-enough” model.

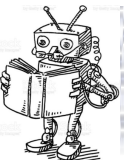
- A non trivial and usually difficult task
- Let's avoid it!

Forecast Combinations

- Instead of looking for the “best-model”, combine diverse predictions
- Simple combinations show accuracy benefits

How to get diverse predictions?

Very bad forecasts will harm the combined predictions



Temporal Hierarchies in a Nutshell

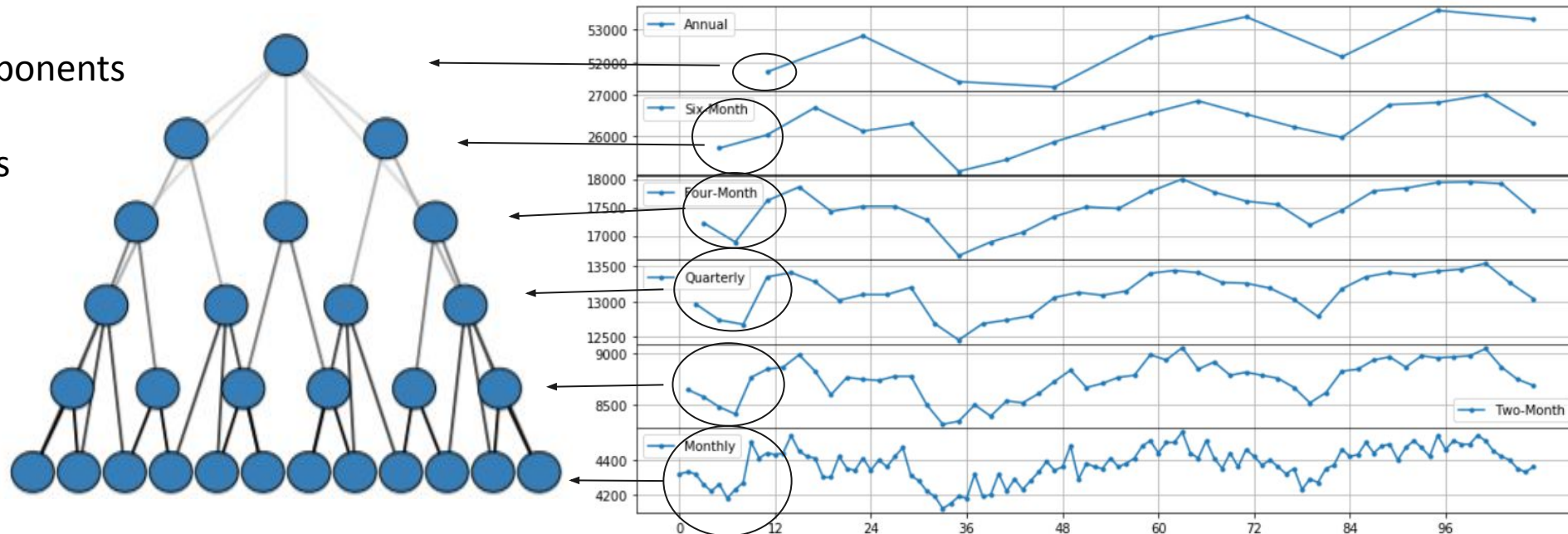
Introduction to Temporal Hierarchies

How to get diverse predictions?

Change your viewpoint through Temporal Aggregations

Temporal Aggregations:

- Filter high frequency components (seasonality)
- Boost high frequency ones (trend)
- Reveal hidden structural information



Temporal Hierarchies in a Nutshell

Introduction to Temporal Hierarchies

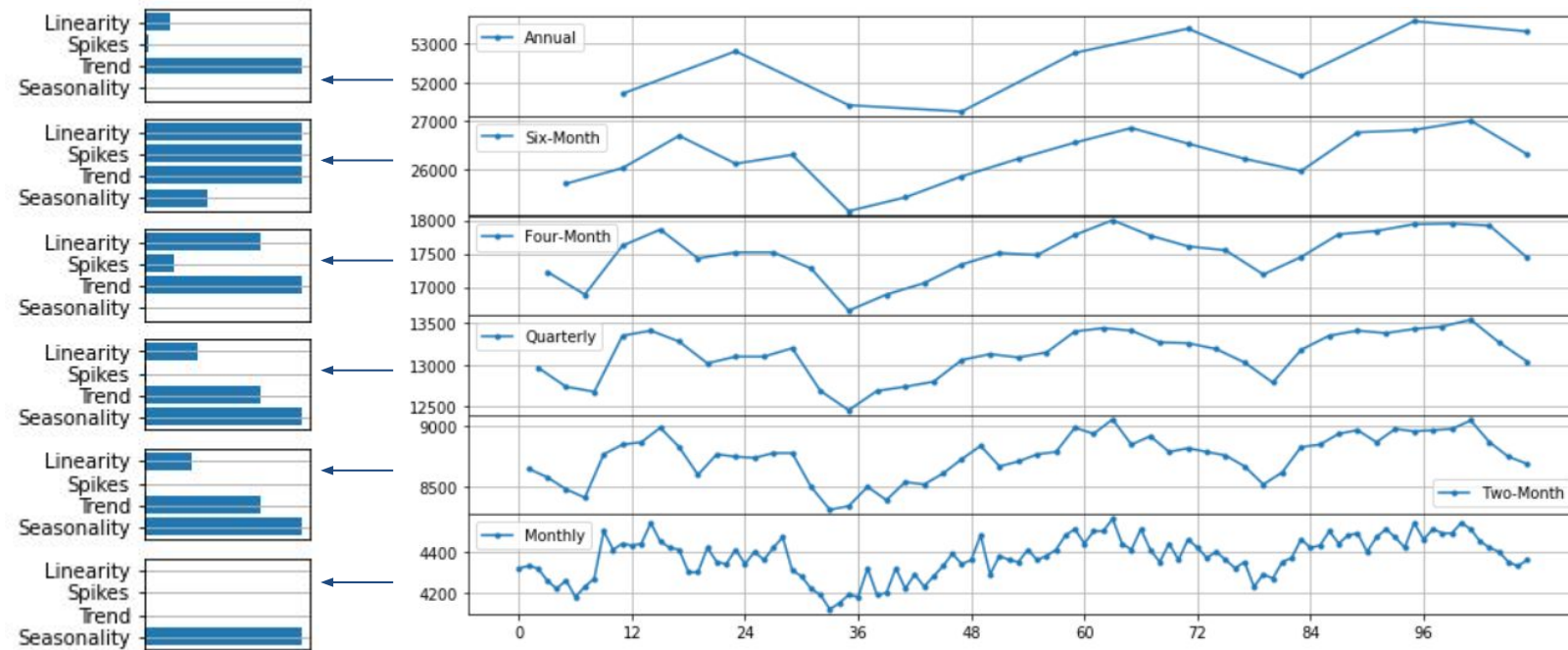
How to get diverse predictions?

Change your viewpoint through Temporal Aggregations

The Diversity Among Levels

Features are strengthen/weaken at each level

What if we combine information from each level into a single forecast?



Features/components estimated using tsfeatures in R

- Spikes measures spikiness. Measured as the variance of the leave-one-out variances of the remainder component after STL
- Linearity is measured based on the coefficients of an orthogonal quadratic regression

Temporal Hierarchies in a Nutshell

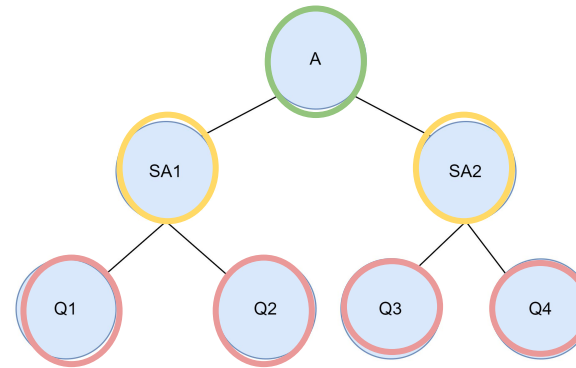
How to Forecast with Temporal Hierarchies (THieF)

Hierarchical Forecasting.

- Bottom up
- Top down -> Biased

Temporal Hierarchy Forecasts (THieF)

1. Generate Base Forecasts $\hat{\mathbf{y}}_h$
2. Reconcile (Just a fancy word for combine) into the bottom level
3. Reconstruct the entire hierarchy



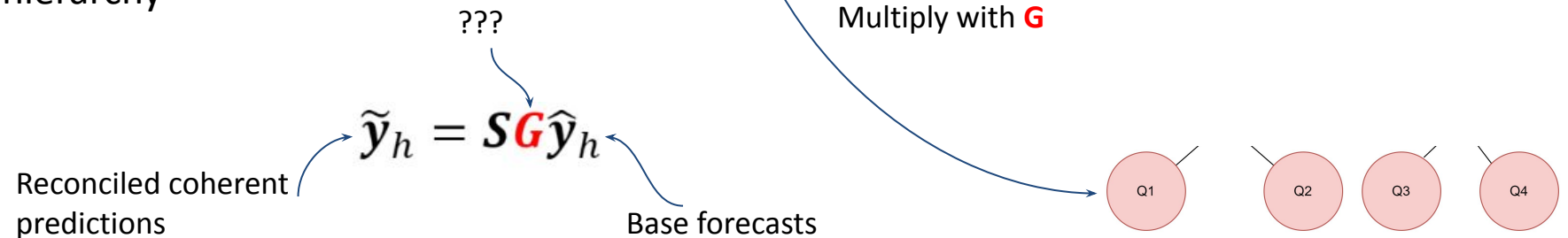
$$\mathbf{b} = (Q1, Q2, Q3, Q4)$$

$$\mathbf{y} = (A, SA1, SA2, \mathbf{b})$$

$$\mathbf{y} = \mathbf{S}\mathbf{b}$$

$$\mathbf{S} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \mathbf{I}_m \end{bmatrix}$$

Top level
Middle level(s)
Bottom level



Temporal Hierarchies in a Nutshell

How to Forecast with Temporal Hierarchies (THieF)

Hierarchical Forecasting.

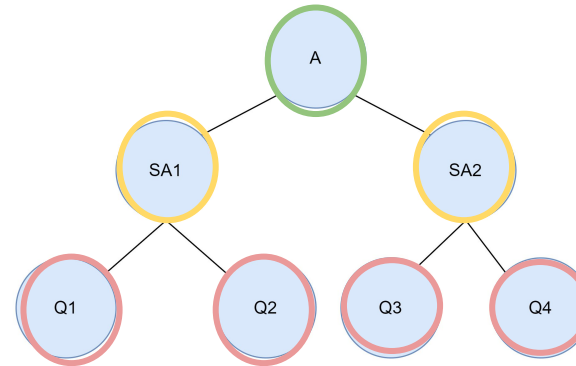
- Bottom up
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Temporal Hierarchy Forecasts (THieF)

1. Generate Base Forecasts $\hat{y}_h$
2. Reconcile (Just a fancy word for combine) into the bottom level
3. Reconstruct the entire hierarchy

Reconciled coherent predictions $\tilde{y}_h = S G \hat{y}_h$ Base forecasts

???



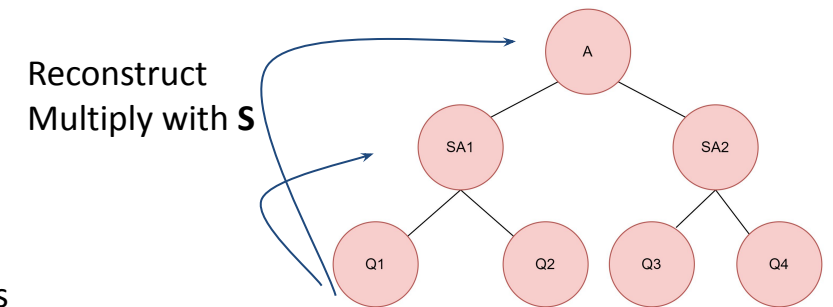
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Top level
Middle level(s)
Bottom level



Temporal Hierarchies in a Nutshell

Estimate **G**

Optimal Reconciliation Matrix G : $\longrightarrow G = (S'W_h^{-1}S)^{-1}S'W_h^{-1}$
(in terms of least squares)

W_h is the positive definite covariance matrix of base forecast errors.

Problem: Estimating W_h in practise is very challenging.

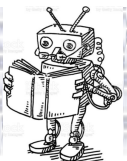
Solution: Again, avoid the problem and rely on approximations

Two diagonal estimators which will be useful later:

1. **Structural Scaling:** A diagonal matrix containing the number of errors contributing to each level.
2. **Variance Scaling:** A diagonal matrix with the in-sample one-step-ahead error variance of each level.

Possible Limitations:

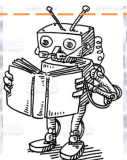
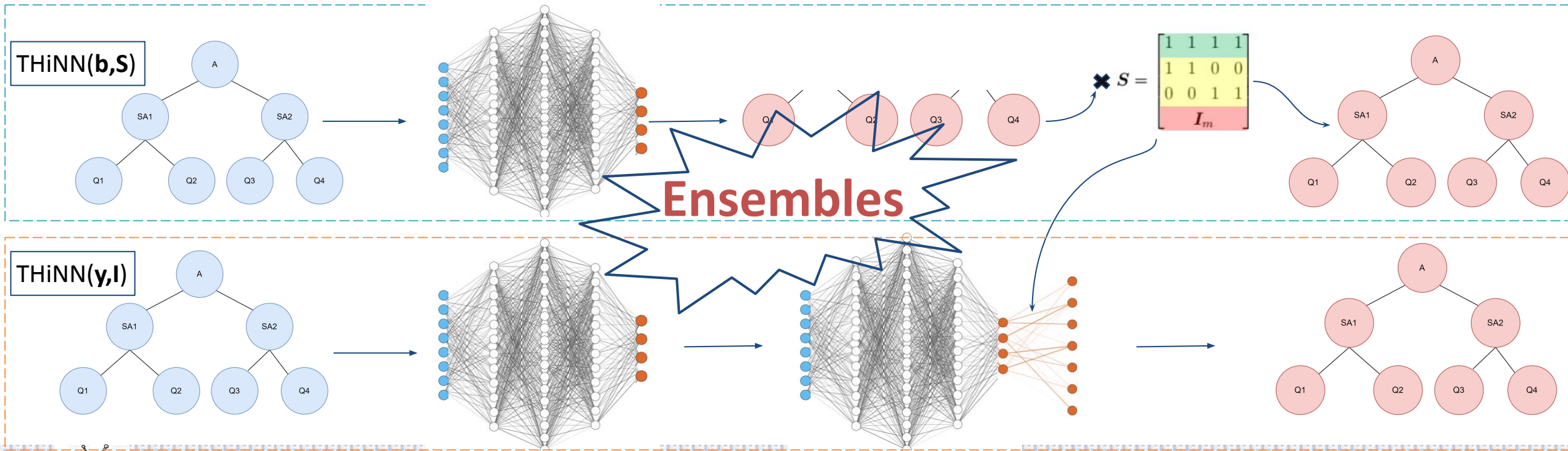
1. Strict Linearity $\longrightarrow$ limited capacity
2. Partial Covariance Estimation $\longrightarrow$ partial consideration of relationships between levels
3. Assumptions $\longrightarrow$ each estimators relies on a number of assumptions



Temporal Hierarchy Neural Network (THiNN) A Non-Linear Data-Driven Reconciler

Approximate **G** with Temporal Hierarchy Neural Network (THiNN)

- NNs: Non-Linear function approximators with success in numerous tasks
- Relax classical restrictions
- Preference over classical ML (Gradient Boosted Decision Trees) due to architectural freedom.



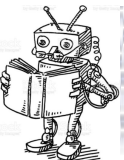
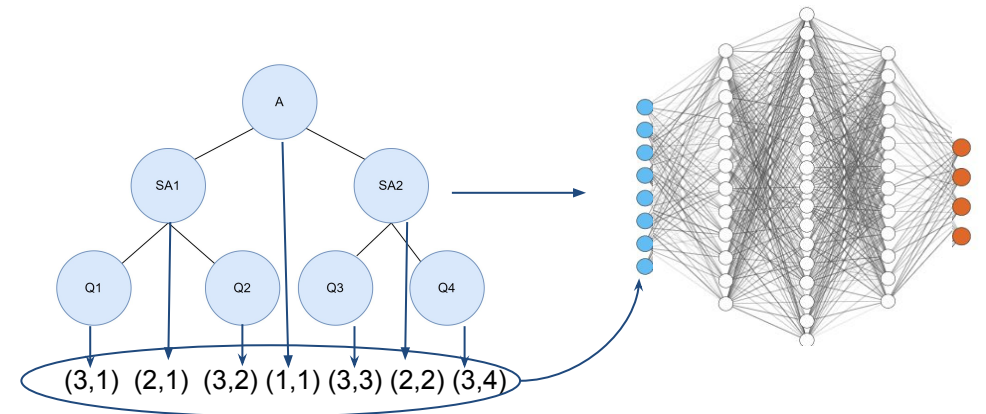
Temporal Hierarchy Neural Network (THiNN) Positional Information

Limitation of Feed Forward NNs:

- Input is random flatten values
- No positional information
- No level-specific information
- Hard time understanding values

Positional Encoding :

- Inspired by Transformers
- Described the location (level) and the position (time-step) of the forecast.
- Pair of type (level, time-step) -> fails on large hierarchies
- Pairs of (cos,sin)
- One-hot-encoded values



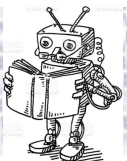
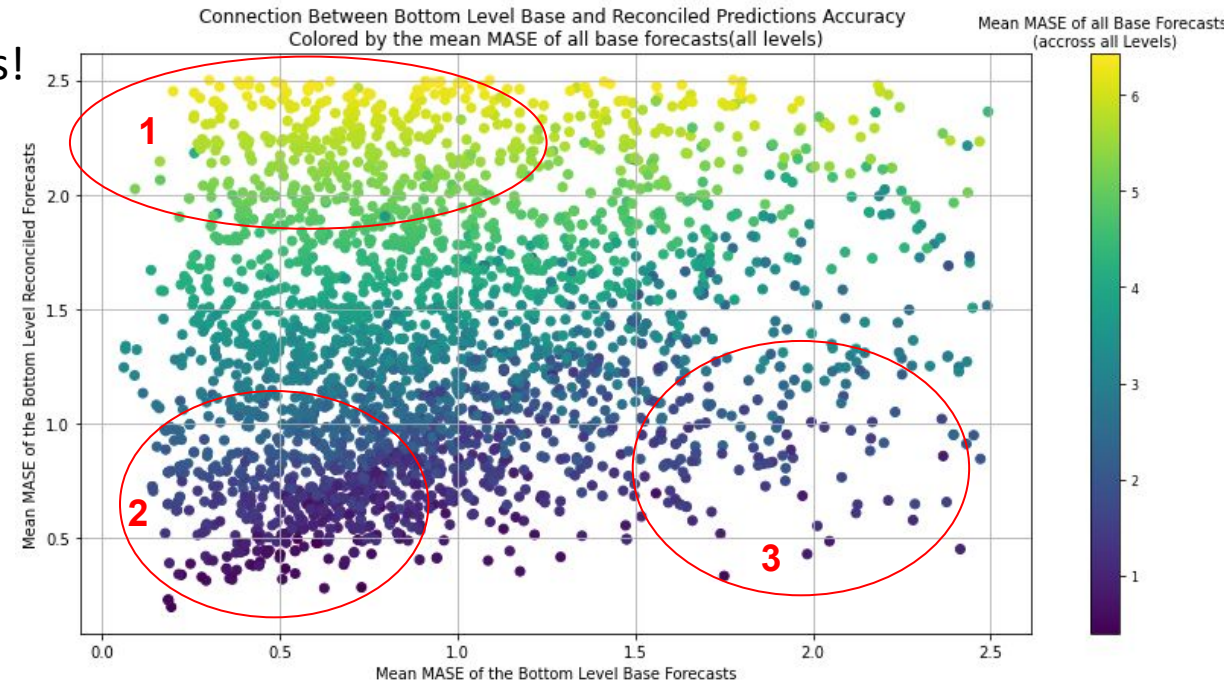
Temporal Hierarchy Neural Network (THiNN) Connection between Base and Reconciled Forecasts

Inaccurate initial forecasts can lead to **bad** forecast combinations!
So, **Bad** base forecasts can **negatively** affect reconciliation?

- Measure MASE of **bottom level** (MASE_b) base forecasts (ETS)
- Measure MASE of **all** (MASE_tot) base forecasts (ETS)
- Measure MASE of **bottom level** (MASE_rec) reconciled
 - Use Structural Scaling (no parametric uncertainty)
 - Accurate reconciled bottom level => Accurate reconciled hierarchy (coherence)

Results:

1. **Low** MASE_B & **High** MASE_tot => **High** MASE_rec
2. **Low** MASE_B & **Low** MASE_tot => **Low** MASE_rec
3. **High** MASE_B & **Low** MASE_tot => **Low** MASE_rec



Pritularga, K.F., Svetunkov, I., Kourentzes, N., 2021. Stochastic coherency in forecast reconciliation. International Journal of Production Economics 240, 108221. <https://doi.org/10.1016/j.ijpe.2021.108221>

Kourentzes, Nikolaos & Barrow, Devon & Petropoulos, Fotios. (2018). Another look at forecast selection and combination: Evidence from forecast pooling. International Journal of Production Economics. 209. 10.1016/j.ijpe.2018.05.019.

Temporal Hierarchy Neural Network (THiNN) Base Forecasts Accuracy “Hints”

Insights:

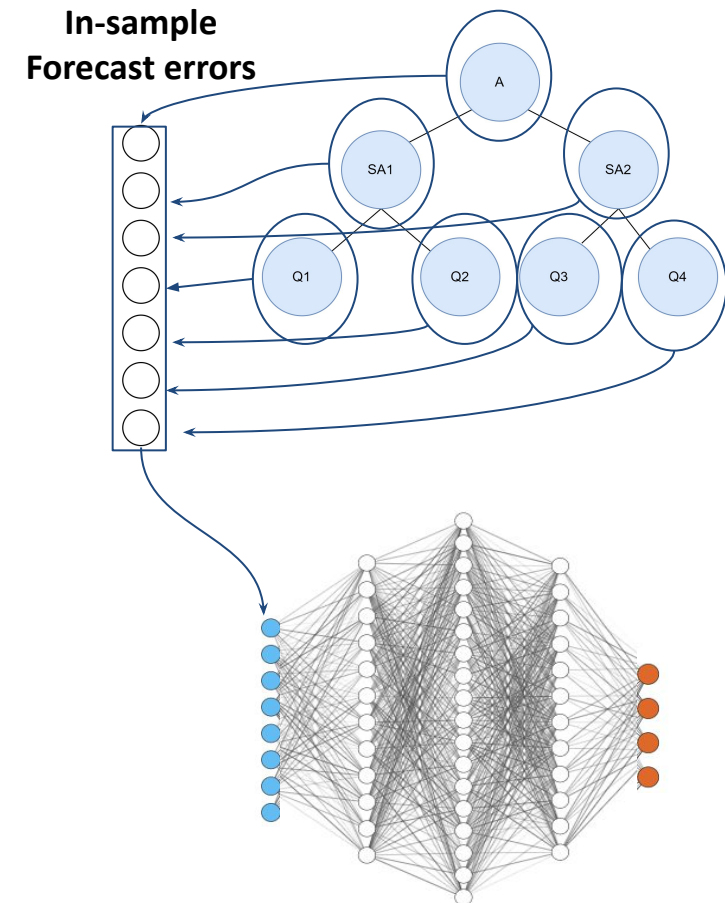
- Having one/two accurate base forecasts does not guarantee good reconciliation
- Bad forecasts drag reconciled predictions outside the accurate subspace
- Problems with model misspecifications on base forecasts (eg ETS for all base forecasts on “difficult” frequencies?)

Ideally THiNN should focus on more accurate base forecasts.
Not possible.

Assumption: in-sample performance gives an adequate picture for out-of-sample

Include mean in-sample accuracy of each base forecast

- THiNN corrects base forecasts based on their in-sample accuracy
- THiNN is not overly affected by inaccurate base forecasts
- Inaccurate base forecasts will be dragged “more” into the accurate subspace



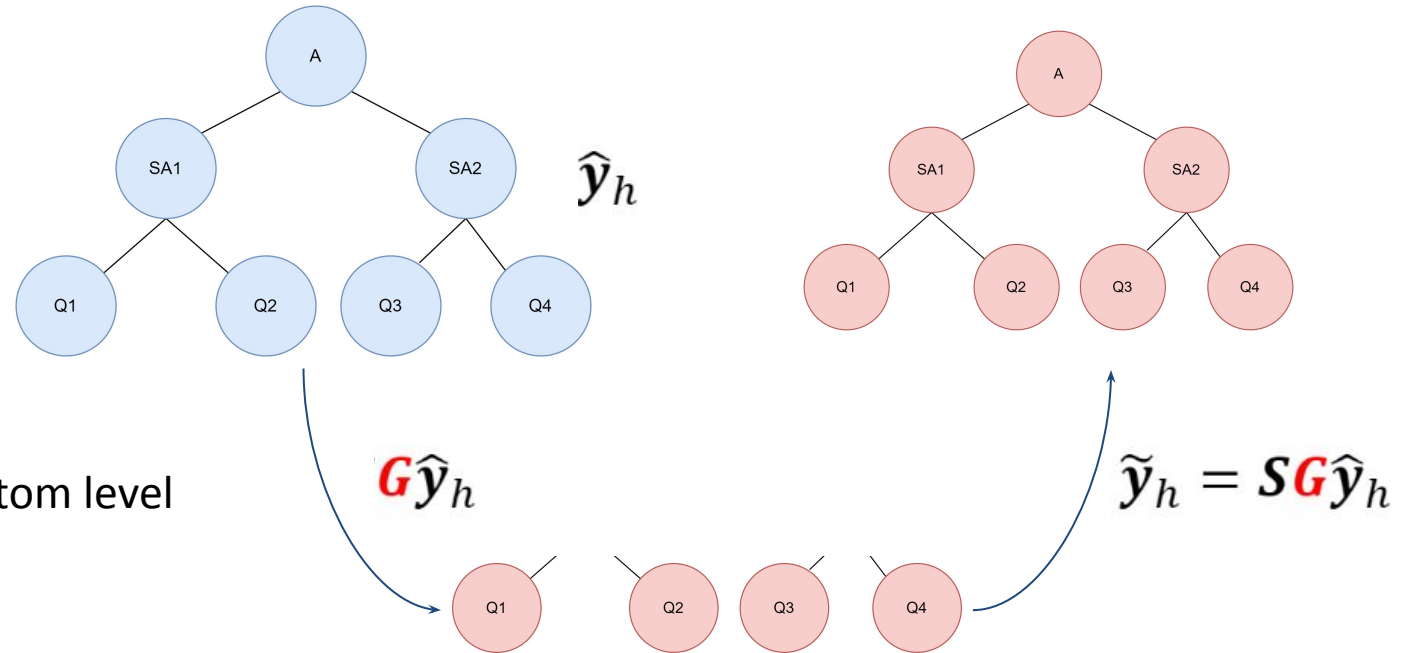
Temporal Hierarchy Neural Network (THiNN) Training the models

Let's review THieF:

1. Get base forecasts for each level
2. Reconcile at the bottom level
3. Reconstruct the hierarchy with bottom-up

Bottom-up:

- Unbiased
- Ensures coherence
- Good hierarchical results given accurate bottom level



Temporal Hierarchy Neural Network (THiNN) Training the models

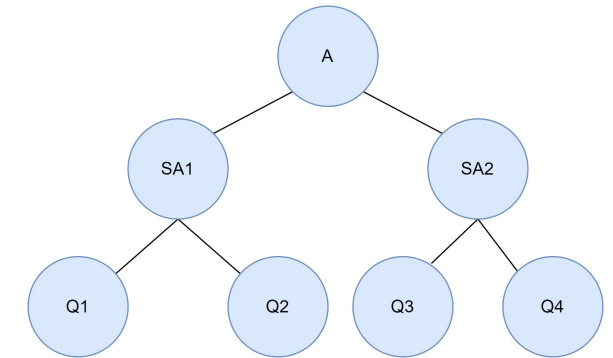
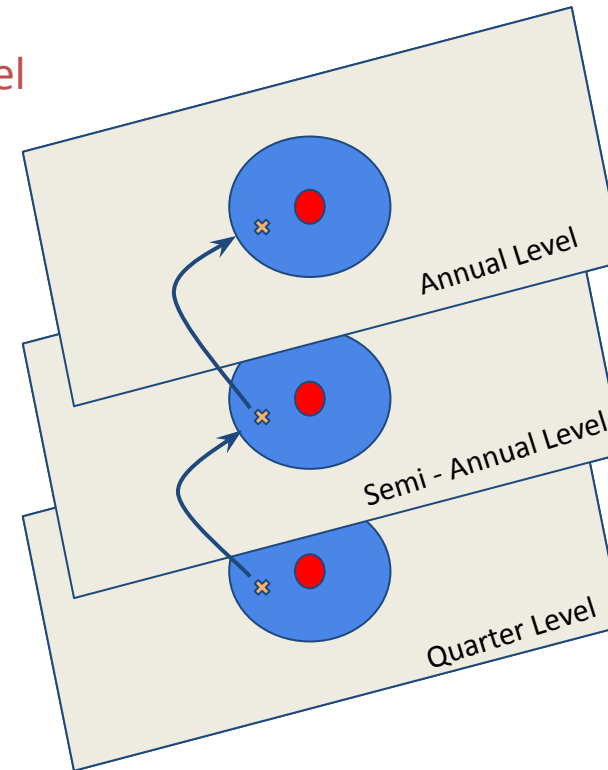
Bottom-up:

- Unbiased
- Ensures coherence
- Good hierarchical results given accurate bottom level

Basic THiNN:

- Gets a set of base forecasts
- Tunes weights to reconcile on bottom level (THieF style)
- Through coherence constraints builds the complete hierarchy

If bottom level is accurate = whole hierarchy is accurate



Properties

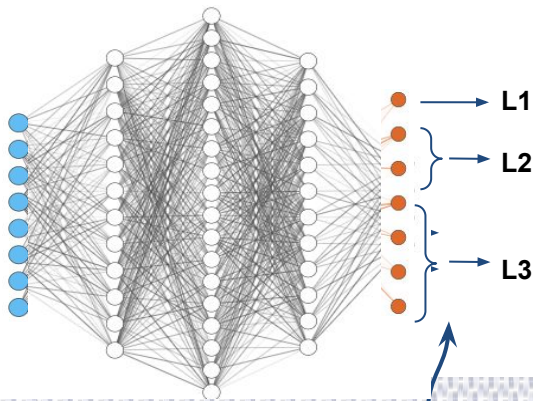
- Focus is on bottom level
- We rely for our reconciliation on bottom level (1-step)
- If it does not work we lose the reconciliation

Temporal Hierarchy Neural Network (THiNN) Training the models

Multi - Level Reconciliation

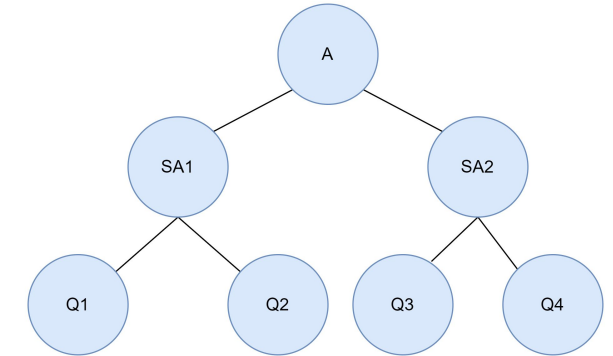
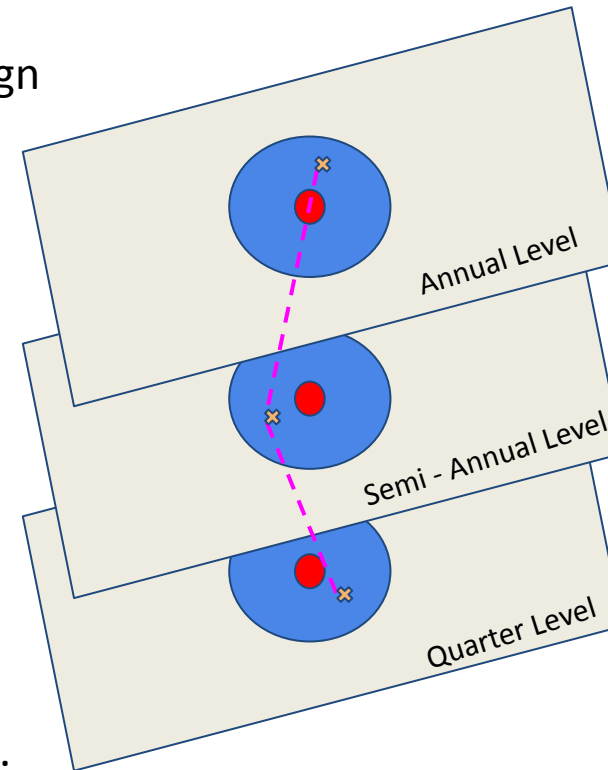
- Instead of tuning weights to only reconcile on bottom level
- Simultaneously optimize on every level and then align
- Objectives of reconciling at each level are identical.
- Shared parameters drags all predictions

Multi - Task Learning



Every Level has its own
loss function

Limitation:
Predictions are not coherent.



Glossary:

- ⊗ Base forecasts
- ⊗ Reconciled Forecast
- - Shared Parameters

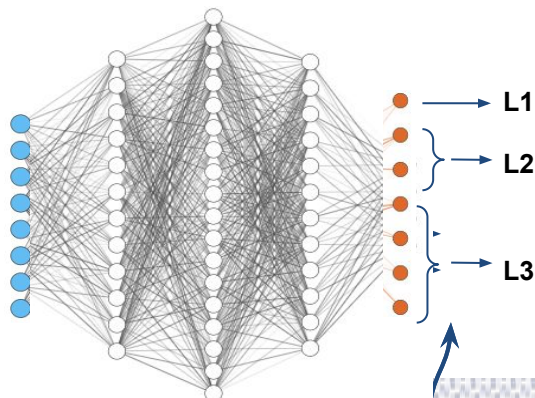
Reconciliation Level: Annual
Steps: i steps to the right

Temporal Hierarchy Neural Network (THiNN) Training the models

Multi - Level Reconciliation

- Instead of tuning weights to only reconcile on bottom level
- Simultaneously optimize on every level and then align
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Multi - Task Learning



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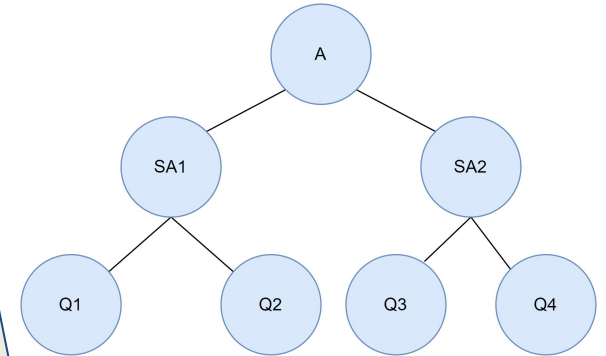
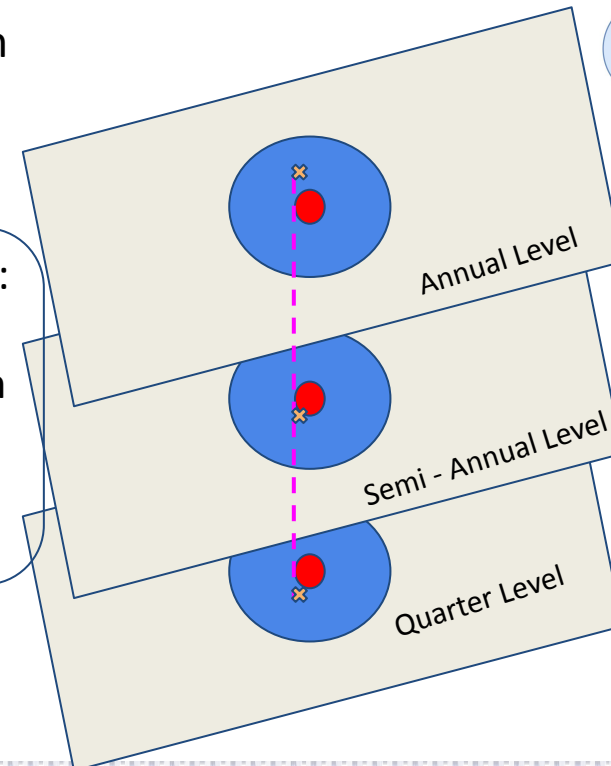


G on Structural Scaling:

- Fixed
- Equal to errors on each level
- Benefits accurate base forecasts

Limitation:

Predictions are not coherent.



Glossary:

- ⊗ Base forecasts
- ⊗ Reconciled Forecast
- Shared Parameters

Reconciliation Level: Structural Scaling
Steps: ALIGNED

Temporal Hierarchy Neural Network (THiNN) A Quick Experiment

Accuracy benefits from each addition

- **Data:** 10.000 monthly time series from M4 (random selection)
Last 12 months kept as test set
- **Accuracy Metric:** Mean Absolute Scaled Error (MASE)
- **Base Forecasts:** ETS
- **Benchmarks:** Structural Scaling

Results:

- NN reconciles **outperform benchmarks** in general
- THiN(y,I) **is not** performing well
- Positional information **not helpful** → misused??
- In-Sample errors **show benefits**
- THiNN - MT and THiNN(y,S) **perform the best**

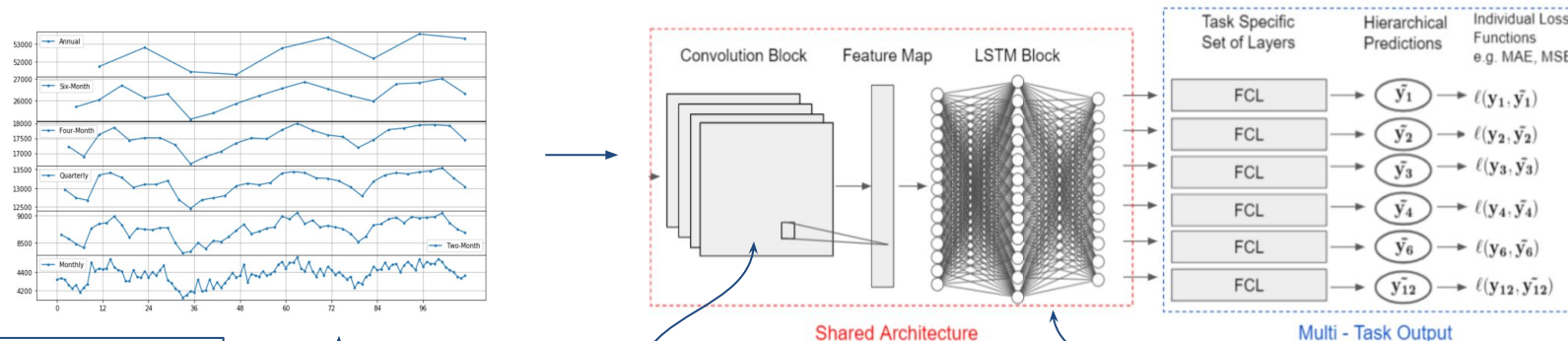
Model	Bottom Level MASE	Complete Hierarchy MASE
Base Forecasts	1.1861	0.9314
Structural Scalling	1,1772	0.91487
THiN(b,S)	1.1744	0.86714
THiN(y,I)	1.3524	0.9974
THiN - MT	1.1752	0.8599
THiN(b,S) + Pos	1.7952	1.2145
THiN(y,I) + Pos	1.6222	1.594
THiN - MT + Pos	1.6612	1.2138
THiN(b,S) + Errors	1.1651	0.8554
THiN(y,I) + Errors	1.5021	1.1102
THiN - MT + Errors	1.1616	0.8334

End-to-End Architectures DeepTHief

Deep Temporal Hierarchies Forecasting (DeepTHief)

- End-to-end learning
 - Single-Model $\longrightarrow$ Base Forecasts + Reconcilers is computationally heavy
 - Shared parameters $\longrightarrow$ End-to-end architectures perform better in such occasions
- All about representation manipulation
- Novel, informative way to represent a single time series

$\longrightarrow$ Higher capacity models with
Better understanding of time series features

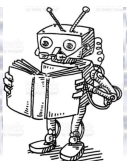


Input Hierarchy
Unlike THief has a not restricted number of temporal levels

CNN Block
Extract Local Relationships among levels

LSTM Block
Models sequential part of time series

Multi - Head Output
Multi-Task module optimized each for each level and then reconciles



End-to-End Architectures Another Quick Experiment

Experimental Setup:

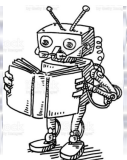
- 12.000 time series from M4
- 12-fold Cross-Validation
- Evaluate Bottom Level & Complete Hierarchy
- Metric MASE

Findings:

1. DeepTHieF **outperforms all methods**
2. THiNN reconcilers perform better than benchmarks
3. THiNN(y,l) performs the best among reconcilers

Table 4: Mean and median MASE results for global training

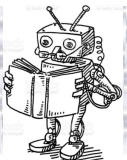
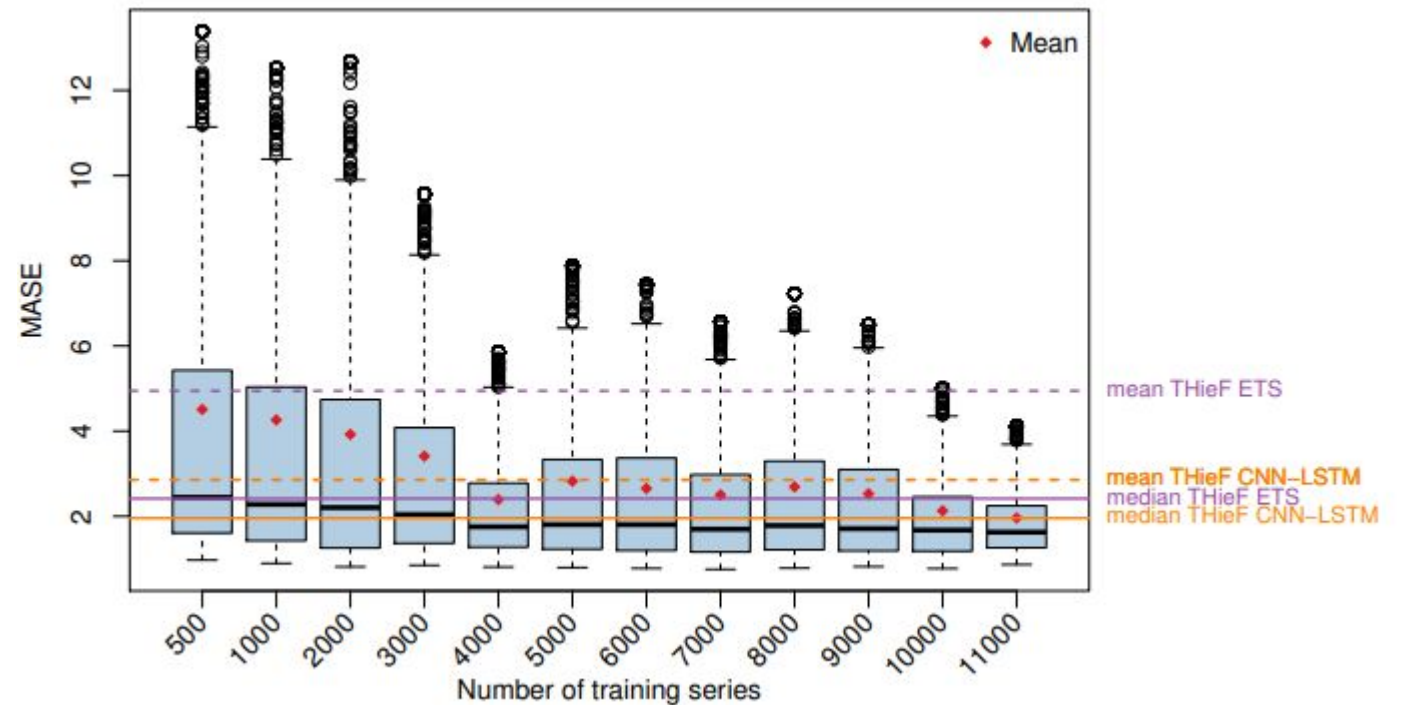
Forecast	Original frequency		Complete Hierarchy	
	Mean	Median	Mean	Median
Base				
ETS	2.6267	<u>1.2742</u>	0.3485	0.2095
Reconciliation & ETS				
Structural	2.6590	1.2973	0.3223	0.1898
THiNN($\hat{b}, S$)	2.6374	1.2823	0.3066	0.1791
THiNN($\hat{y}, I$)	<u>2.6246</u>	1.2663	<u>0.3040</u>	<u>0.1766</u>
THiNN($\hat{b}, \hat{w}$)	2.6294	<u>1.2649</u>	0.3091	0.1845
THiNN($\hat{y}, \hat{w}$)	2.7352	1.3361	0.3173	0.1848
DeepTHieF	1.5479	1.0616	0.2077	0.1659



End-to-End Architectures DeepTHieF Data Requirements

Requirements for Global Training:

- Global DL Forecasting models performance increases linearly to data size
- Still DeepTHieF shows a robust performance on a reduced number of time series
- THieF has an advantage on a small number of time series.



End-to-End Architectures Prediction Intervals (PIs)

THieF by construction is optimized for **Point Forecast Reconciliation**.

Approaches in Literature for Reconciling PIs

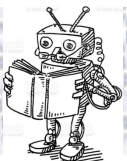
- Naive approach: Reconcile lower and upper prediction interval
- Amazon: Sample from a distribution and map to the coherent subspace
- Empirical Methods

The Flexibility of NNs

- NNs can be optimized on producing PIs
- Appropriate Loss Function
- THiNN can also output quantiles

Examples include but not limited

1. Interval score
 - a. Output Upper & Lower Interval
 - b. Can be combined with MAE -> similar scales
2. Probabilistic
 - a. Assume a distribution
 - b. Predict the parameters (eg μ , σ on Gaussian)



Theodosiou, Filotas & Kourntzes, Nikolaos. (2021). Deep Learning Temporal Hierarchies for Interval Forecasts. International Conference of AI in Finance. Time Series Workshop

Syama Sundar Rangapuram, Lucien D Werner, Konstantinos Benidis, Pedro Mercado, Jan Gasthaus, and Tim Januschowski. 2021. End-to-End Learning of Coherent Probabilistic Forecasts for Hierarchical Time Series. In Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)

Conclusions & Future of NNs and Temporal Hierarchies

- THieF's success is based on combining diverse information
- Different viewpoints produce different features
- THieF struggles when base forecasts are very inaccurate.

We proposed THiNN -> A Data-Driven Non-Linear Reconciler based on THieF

- Positional Information
- Base forecasts accuracy
- Multi-task Learning

DeepTHieF & End-to-End Architectures

- Novel Input representation based on Temporal Hierarchies
- Single NN does not work at each task (not in forecasting)
- Change how global NNs view time series

Forecasting with Deep Temporal Hierarchies

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